

CHARMAINE GUO

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EDUCATION

Northwestern University

B.S. in Materials Science and Engineering, Double Major in Data Science

June 2028

SKILLS

Computer Languages

Soft Skills

Hard Skills

Organizational Tools

R, SQL, MongoDB, Python, Java, MATLAB, HTML, CSS

Project Management, Teamwork, Time-Management, Adaptability

Research Methods, Experimental Design, Data Analysis, Metallography

Microsoft Excel, Google Drive, Github

EXPERIENCE

Student Tech Aide

The Garage

September 2025 – Present

Evanston, IL

- Manage makerspace tools and inventory while supporting prototyping for student entrepreneurs across manufacturing platforms.
- Translate startup project requirements into safe fabrication workflows, enabling rapid prototyping within Northwestern's entrepreneurship hub.

Metallurgy Intern

General Motors (GM)

June 2026 – Sept 2026

Toledo, OH

- Designed and analyzed 15+ carburizing and quench experiments to improve case-depth control while maintaining surface hardness.
- Characterized 70+ treated gear samples through metallography, including puck preparation, polishing, etching, and microscopy to characterize grain size, and martensite and retained austenite vs. spec.
- Analyzed 70+ gears in Excel, linking load position to hardness, case depth, microstructure variation.

Undergraduate Researcher — Schuh Group

Northwestern University

Jan 2026 – March 2026

Evanston, IL

- Synthesized zircon (ZrSiO_4) coatings on micron-scale sand particles using the Pechini method, leveraging polymerizable precursors to ensure precise stoichiometric control and chemical homogeneity.
- Refined sol-gel chemistry to stabilize resin precursors and improve uniform particle encapsulation.

Quality Intern

Littelfuse

June 2025 – August 2025

Beverly, MA

- Built centralized repository with Power Query, automating ETL for 300+ legacy quality reports.
- Developed multi-parameter search tool, enabling instant queries by part, package, and leadframe attributes.
- Optimized engineering report search time from several hours to under one minute.

Undergraduate Researcher — SPln Lab

Northwestern University

October 2024 – June 2025

Evanston, IL

- Fabricated cellulose-acrylic films to investigate how polymer molecular-weight distribution affected UV-curing and dye-free structural color.
- Analyzed MATLAB data to quantify effects of molecular-weight composition and cellulose concentration on crystal arrangement and color response.

PROJECTS

Pallet Tracking

Siemens STEP 7 (LAD), TIA Portal, Allen-Bradley Studio 5000/Logix Designer

Independent Project at General Motors

Aug - Sept 2026

- Recreated Allen-Bradley pallet-identification and counting logic in Siemens STEP 7 using an FB/instance DB, a 256-bit pallet-history array, RFID ID mapping, daily reset logic, and one-shot scan handling.
- Built a laser-triggered simulation with HMI-adjustable RFID IDs to validate.

Skynet System

Onshape, Arduino IDE, ESP32

Team Project for Lincoln Park Zoo

January - March 2025

- Designed and prototyped a remote zoo-access system for Lincoln Park Zoo, culminating in a remotely controlled, zipline-mounted camera that provided users with a live video feed.

EXTRACURRICULAR ACTIVITIES

Low Voltage Team

Northwestern Formula SAE (NUFSAE)

October 2024 – Present

- Designed Power Distribution Module PCB in Altium, owning full schematic capture and layout.
- Writing wiring harness specs: connector selection, pinout mapping, and signal/safety routing.
- Leading technical onboarding for new members by delegating design tasks and conducting training sessions on RapidHarness and industry-standard documentation.